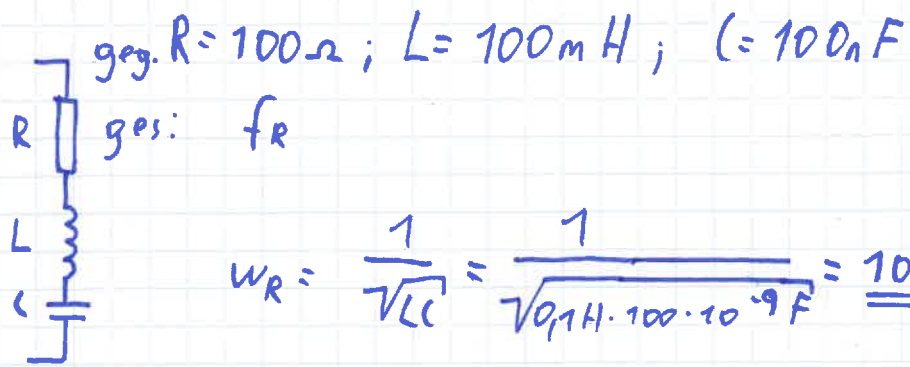


Bsp 1

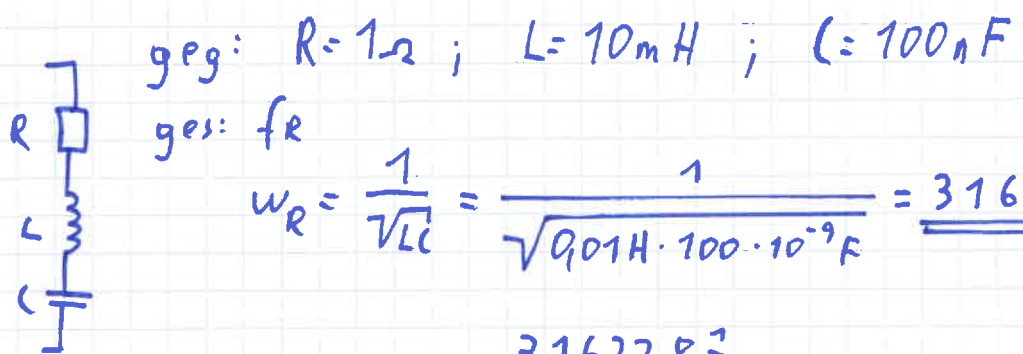


$$\omega_R = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0,1 \text{ H} \cdot 100 \cdot 10^{-9} \text{ F}}} = \underline{\underline{10 \text{ kHz}}}$$

$$\omega_R = 2\pi f_R$$

$$f_R = \frac{\omega_R}{2\pi} = \frac{10 \text{ kHz}}{2\pi} = \underline{\underline{1,592 \text{ kHz}}}$$

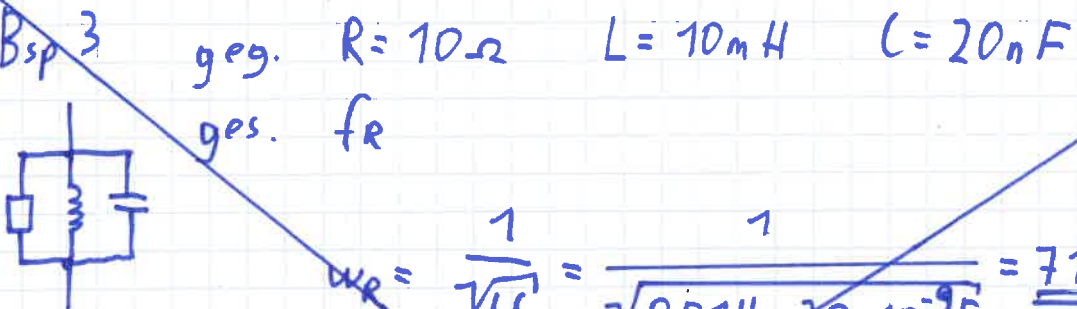
Bsp 2



$$\omega_R = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0,01 \text{ H} \cdot 100 \cdot 10^{-9} \text{ F}}} = \underline{\underline{31622,8 \frac{1}{\text{s}}}}$$

$$f_R = \frac{\omega_R}{2\pi} = \frac{31622,8 \frac{1}{\text{s}}}{2\pi} = \underline{\underline{5032,92 \text{ Hz}}}$$

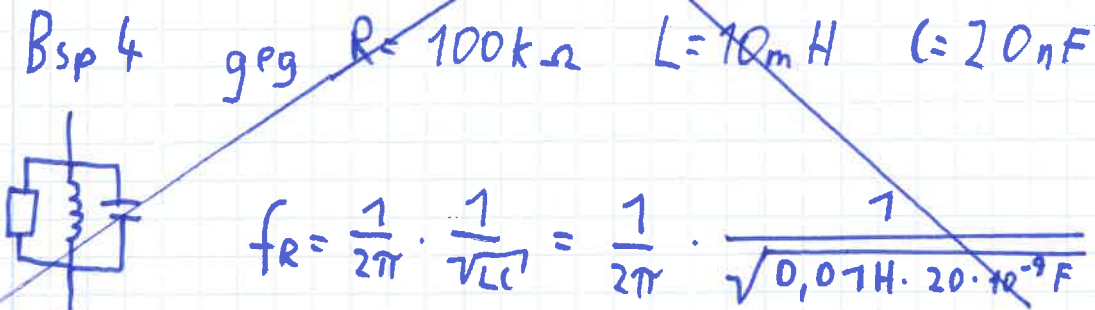
~~Bsp 3~~



~~$$\omega_R = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{0,01 \text{ H} \cdot 20 \cdot 10^{-9} \text{ F}}} = \underline{\underline{715542 \frac{1}{\text{s}}}}}$$~~

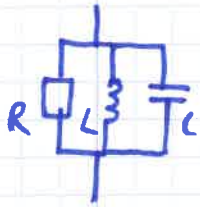
~~$$f_R = \frac{\omega_R}{2\pi} = \frac{715542 \frac{1}{\text{s}}}{2\pi} = \underline{\underline{113882 \text{ Hz}}}$$~~

~~Bsp 4~~



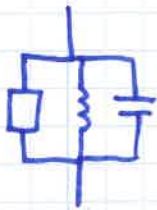
~~$$f_R = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{LC}} = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{0,01 \text{ H} \cdot 20 \cdot 10^{-9} \text{ F}}} =$$~~

Bsp 3 geg $R = 10 \Omega$ $L = 10 \text{ mH}$ ($C = 20 \text{ nF}$)
 ges f_R



$$f_R = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{LC}} = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{0,01 \text{ H} \cdot 20 \cdot 10^{-9} \text{ F}}} = \underline{\underline{11254 \frac{1}{\text{s}}}}$$

Bsp 4 geg $R = 100 \text{ k}\Omega$ $L = 10 \text{ mH}$ ($C = 20 \text{ nF}$)



$$f_R = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{LC}} = \frac{1}{2\pi} \cdot \frac{1}{\sqrt{0,01 \text{ H} \cdot 20 \cdot 10^{-9} \text{ F}}} = \underline{\underline{11254 \frac{1}{\text{s}}}}$$

wie Bsp 3, da f_R unabhängig von R ist! ($\frac{1}{\text{s}} = \text{Hz}$)

Bsp 5 ges: Herleitung f_R



$$\begin{aligned} \underline{Z_g} &= R + j\omega L + \frac{1}{j\omega C} = R + j\omega L + \frac{1}{j\omega C} \cdot \frac{j}{j} = \\ &= R + j\omega L - \frac{j}{\omega C} = R + jX \end{aligned}$$

ω_R bei $X = 0$:

$$j\omega_R L - \frac{j}{\omega_R C} = 0$$

$$\omega_R L - \frac{1}{\omega_R C} = 0$$

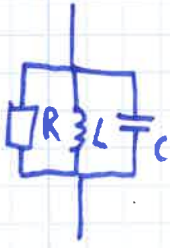
$$\omega_R L = \frac{1}{\omega_R C}$$

$$\omega_R^2 = \frac{1}{LC}$$

$$\underline{\underline{\omega_R = \frac{1}{\sqrt{LC}}}}$$

$$f_R = \frac{\omega_R}{2\pi}$$

Bsp 6 ges. Herleitung f_R



$$Y_0 = \frac{1}{R} + \frac{1}{j\omega L} + \frac{1}{\frac{1}{j\omega C}} = \frac{1}{R} + \frac{1}{j\omega L} + \frac{j\omega C}{1} =$$

$$= \frac{1}{R} + \frac{j}{j} \cdot \frac{1}{j\omega L} + j\omega C = \frac{1}{R} - \frac{j}{\omega L} + j\omega C$$

ω_R bei $B=0$ ($\text{Im}(Y)=0$)

$$-\frac{1}{\omega_R L} + \omega_R C = 0$$

$$\omega_R C = \frac{1}{\omega_R L}$$

$$\omega_R^2 = \frac{1}{LC}$$

$$\underline{\underline{\omega_R = \frac{1}{\sqrt{LC}}}}$$

$$f_R = \frac{\omega_R}{2\pi}$$

Bsp 7 ges. $Q = \frac{U_L}{U} = \frac{U_C}{U}$ geg. R, L, C



$$Q = \frac{1}{R} \cdot \sqrt{\frac{L}{C}} = \frac{1}{12 \cdot 2} \cdot \sqrt{\frac{0,0354}{1 \cdot 10^{-6} \text{ F}}} = \underline{\underline{15,6}}$$

$$U_{C_{\text{eff}}} = Q \cdot U_{W_{\text{eff}}} = 15,6 \cdot 20 \text{ V} = \underline{\underline{312 \text{ V}}}$$

Bsp 8

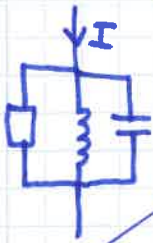


geg R, L, C

ges $Q = \frac{U_L}{U} = \frac{U_C}{U}$

$$Q = \frac{1}{R} \cdot \sqrt{\frac{L}{C}} = \frac{1}{100\Omega} \cdot \sqrt{\frac{0,1H}{100 \cdot 10^{-9}F}} = \underline{\underline{10}}$$

~~Bsp 9~~



~~geg R, L, C~~

~~ges $Q = \frac{I_C}{I} = \frac{I_L}{I}$~~

~~$Q = \frac{1}{R} \cdot \sqrt{\frac{C}{L}} =$~~

~~Bsp 9~~

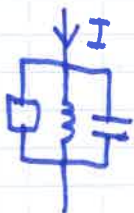


~~geg R, L, C~~

~~ges $Q = \frac{I_C}{I} = \frac{I_L}{I}$~~

~~$Q = \frac{1}{R} \cdot \sqrt{\frac{C}{L}} =$~~

Bsp 9



geg: R, L, C

ges: $Q = \frac{I_C}{I} = \frac{I_L}{I}$

$$\begin{cases} R = 10\Omega \\ L = 10mH \\ C = 20nF \end{cases}$$

$$Q = R \cdot \sqrt{\frac{C}{L}} = \underline{\underline{0,014}}$$

Bei starker Dämpfung kann die Güte bzw. die Resonanzüberhöhung kleiner 1 sein!

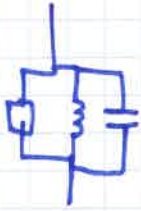
Bsp 10:

geg $L = 40 \text{ mH}$

$G = 1,2 \text{ mS}$

$C = 2,2 \text{ pF}$

$U = 10 \text{ V}$



ges: alle U , alle I , alle P/Q

$$Q = R \cdot \sqrt{\frac{C}{L}} = \frac{1}{G} \cdot \sqrt{\frac{C}{L}} = \frac{1}{1,2 \cdot 10^{-3} \Omega} \cdot \sqrt{\frac{2,2 \cdot 10^{-6} \text{ F}}{0,040 \text{ H}}} = \underline{\underline{6,18}}$$

$I_R = U \cdot G = \underline{\underline{12 \text{ mA}}}$ (bei Resonanz ist $B=0$)

$I_C = I_L = I_R \cdot Q = 12 \text{ mA} \cdot 6,18 = \underline{\underline{74,2 \text{ mA}}}$

$U_R = U_L = U_C = \underline{\underline{10 \text{ V}}}$

$P = U \cdot I = 10 \text{ V} \cdot 12 \text{ mA} = \underline{\underline{0,12 \text{ W}}}$

$Q_C = Q_L = U \cdot I_C = U \cdot I_L = 10 \text{ V} \cdot 74,2 \text{ mA} = \underline{\underline{0,74 \text{ W}}}$

(W = var)

Bsp 11)



$\Rightarrow$

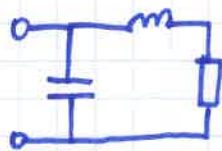


ges: C

$R_{NEU} = R_R = \frac{L}{R \cdot C}$

$C = \frac{L}{R \cdot R_R} = \frac{0,1 \text{ H}}{50 \Omega \cdot 70 \Omega} = \underline{\underline{29 \text{ pF}}}$

Bsp 12: a) ges Y, ω_R, f_R



$\rightarrow LR$ (ohet parallel $\rightarrow Y$!)

$$Y_G = Y_C + Y_{LR}$$

$$Y_C = \frac{1}{\frac{1}{j\omega C}} = j\omega C$$

$$Y_{LR} = \frac{1}{Z_{LR}} = \frac{1}{R + j\omega L}$$

$$Y_G = j\omega C + \frac{1}{R + j\omega L} = j\omega C + \frac{1}{R + j\omega L} \cdot \underbrace{\frac{R - j\omega L}{R - j\omega L}}_{=1} =$$

$$= j\omega C + \frac{R - j\omega L}{R^2 - j^2 \omega^2 L^2} = j\omega C + \frac{R - j\omega L}{R^2 + \omega^2 L^2} =$$

$$= \frac{R}{R^2 + \omega^2 L^2} + j \underbrace{\left(\omega C + \frac{-\omega L}{R^2 + \omega^2 L^2} \right)}_{B=0}$$

~~$$\omega C - \frac{\omega L}{R^2 + \omega^2 L^2} = 0 \quad | : \omega$$~~

~~$$C - \frac{L}{R^2 + \omega^2 L^2} = 0$$~~

~~$$C = \frac{1}{\omega R^2 L}$$~~

$$\omega C - \frac{\omega L}{R^2 + \omega^2 L^2} = 0$$

$$\omega C = \frac{\omega L}{R^2 + \omega^2 L^2} \quad | \cdot R^2 + \omega^2 L^2$$

$$\omega C (R^2 + \omega^2 L^2) = \omega L \quad | : \omega$$

$$C (R^2 + \omega^2 L^2) = L$$

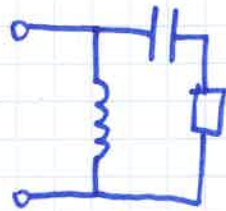
$$\omega^2 L^2 C = L - R^2 C \quad | : L^2 C$$

$$\omega^2 = \frac{L}{L^2 C} - \frac{R^2 C}{L^2 C}$$

$$\omega^2 = \frac{1}{LC} - \frac{R^2 C}{LC \cdot L} \Rightarrow \underline{\underline{\omega = \frac{1}{\sqrt{LC}} \cdot \sqrt{1 - R^2 \frac{C}{L}}}}$$

$$f_R = \frac{\omega_R}{2\pi}$$

12b) ges $\underline{Y}$ ω_R f_R



L & C eher parallel $\rightarrow \underline{Y}$!

$$\underline{Y} = \underline{Y}_L + \underline{Y}_{RC}$$

$$\underline{Y}_L = \frac{1}{j\omega L}$$

$$\underline{Y}_{RC} = \frac{1}{R + \frac{1}{j\omega C}} = \frac{1}{R - \frac{j}{\omega C}}$$

$$\underline{Y} = \frac{1}{j\omega L} + \frac{1}{R - \frac{j}{\omega C}} = \frac{-j}{\omega L} + \frac{1}{R - \frac{j}{\omega C}} \cdot \frac{R + \frac{j}{\omega C}}{R + \frac{j}{\omega C}} =$$

$$= \frac{-j}{\omega L} + \frac{R + \frac{j}{\omega C}}{R^2 + \frac{1}{\omega^2 C^2}} = \frac{R}{R^2 + \frac{1}{\omega^2 C^2}} + j \left(\frac{\frac{1}{\omega C}}{R^2 + \frac{1}{\omega^2 C^2}} - \frac{1}{\omega L} \right)$$

$$\frac{\frac{1}{\omega C}}{R^2 + \frac{1}{\omega^2 C^2}} - \frac{1}{\omega L} = 0$$

$$\frac{\frac{1}{\omega C}}{R^2 + \frac{1}{\omega^2 C^2}} \cdot \frac{\omega^2 C^2}{\omega^2 C^2} = \frac{1}{\omega L}$$

$$\frac{\omega C}{R^2 \omega^2 C^2 + 1} = \frac{1}{\omega L}$$

$$\omega^2 L C = R^2 \omega^2 C^2 + 1$$

$$\omega^2 L C - \omega^2 R^2 C^2 = 1$$

$$\omega^2 (L C - \omega^2 R^2 C^2) = 1 \quad | : L C$$

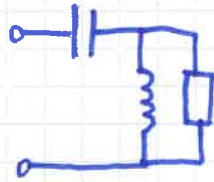
$$\omega^2 \left(1 - R^2 \frac{C}{L} \right) = \frac{1}{L C}$$

$$\omega^2 = \frac{1}{L C} \cdot \frac{1}{1 - R^2 \frac{C}{L}}$$

$$\omega_R = \frac{1}{\sqrt{L C}} \cdot \frac{1}{\sqrt{1 - R^2 \frac{C}{L}}}$$

$$f_R = \frac{\omega_R}{2\pi}$$

13 a)



$$Z = \frac{1}{j\omega L} + \frac{1}{\frac{1}{j\omega L} + \frac{1}{R}} = \frac{-j}{\omega L} + \frac{1}{\frac{-j}{\omega L} + \frac{1}{R}}$$

$$= -\frac{j}{\omega L} + \frac{1}{\frac{1}{R} - \frac{j}{\omega L}} \cdot \frac{\frac{1}{R} + \frac{j}{\omega L}}{\frac{1}{R} + \frac{j}{\omega L}} =$$

$$= -\frac{j}{\omega L} + \frac{\frac{1}{R} + \frac{j}{\omega L}}{\left(\frac{1}{R}\right)^2 + \frac{1}{\omega^2 L^2}} =$$

$$= \frac{\frac{1}{R}}{\left(\frac{1}{R}\right)^2 + \frac{1}{\omega^2 L^2}} + j \underbrace{\left(\frac{-1}{\omega L} + \frac{\frac{1}{\omega L}}{\frac{1}{R^2} + \frac{1}{\omega^2 L^2}} \right)}_{X=0}$$

$$-\frac{1}{\omega L} + \frac{\frac{1}{\omega L}}{\frac{1}{R^2} + \frac{1}{\omega^2 L^2}} = 0$$

$$\frac{\frac{1}{\omega L}}{\frac{1}{R^2} + \frac{1}{\omega^2 L^2}} = \frac{1}{\omega L}$$

$$\frac{\frac{1}{\omega L}}{\frac{1}{R^2} + \frac{1}{\omega^2 L^2}} \quad \frac{\omega^2 L^2}{\omega^2 L^2} = \frac{1}{\omega C}$$

$$\frac{\omega L}{\frac{\omega^2 L^2}{R^2} + 1} = \frac{1}{\omega C}$$

$$\frac{\omega L}{\frac{\omega^2 L^2}{R^2} + \frac{R^2}{R^2}} = \frac{1}{\omega C}$$

$$\frac{R^2 \omega L}{\omega^2 L^2 + R^2} = \frac{1}{\omega C}$$

$$R^2 \omega L \omega C = \omega^2 L^2 + R^2$$

$$\omega^2 R^2 L C = \omega^2 L^2 + R^2$$

$$\omega^2 \cdot (R^2 L C - L^2) = R^2$$

$$\omega^2 = \frac{1}{R^2 L C - L^2} \cdot R^2 = \frac{1}{L C - \frac{L^2}{R^2}} = \frac{1}{L C} \cdot \left(\frac{1}{1 - \frac{L}{R^2 C}} \right)$$

$$\omega_R = \frac{1}{\sqrt{L C}} \cdot \sqrt{\frac{1}{1 - \frac{L}{R^2 C}}}$$

$$f_R = \frac{\omega_R}{2\pi}$$

13 b)



$$\underline{Z} = j\omega L + \frac{1}{j\omega C + \frac{1}{R}}$$

$$\underline{Z} = j\omega L + \frac{1}{\frac{1}{R} + j\omega C} \cdot \frac{\frac{1}{R} - j\omega C}{\frac{1}{R} - j\omega C} =$$

$$= j\omega L + \frac{\frac{1}{R} - j\omega C}{\frac{1}{R^2} + \omega^2 C^2} =$$

$$= \frac{\frac{1}{R}}{\frac{1}{R^2} + \omega^2 C^2} + j \left(\omega L - \frac{\omega C}{\frac{1}{R^2} + \omega^2 C^2} \right)$$

$$\omega L = \frac{\omega C}{\frac{1}{R^2} + \omega^2 C^2}$$

$$\begin{array}{l} | \cdot (\frac{1}{R^2} + \omega^2 C^2) \\ | : \omega L \end{array}$$

$$\frac{1}{R^2} + \omega^2 C^2 = \frac{\omega C}{\omega L}$$

$$\omega^2 C^2 = \frac{C}{L} - \frac{1}{R^2}$$

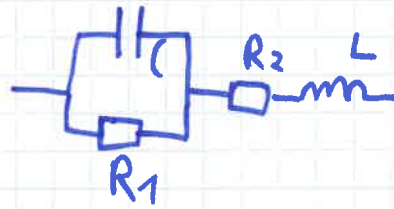
$$\omega^2 = \frac{1}{LC} - \frac{1}{R^2 C^2} = \frac{1}{LC} - \frac{1}{R^2 C^2 L}$$

$$\omega^2 = \frac{1}{LC} \cdot \left(1 - \frac{1}{R^2} \cdot \frac{L}{C} \right)$$

$$\underline{\omega_R} = \frac{1}{\sqrt{LC}} \cdot \sqrt{1 - \frac{1}{R^2} \frac{L}{C}}$$

$$f_R = \frac{\omega_R}{2\pi}$$

Bsp 14)



$$\underline{Z} = R_2 + j\omega L + \frac{1}{\frac{1}{R_1} + j\omega C}$$

wie Bsp 13b)

(R_2 wirkt sich auf f_R nicht aus!)

Bsp 15 ges $U = ?$

geg $A = -3 \text{ dB}_v$

$$\text{dB}_v \rightarrow U_1 = 0,775 \text{ V}$$

$$A = 20 \log\left(\frac{U_2}{U_1}\right)$$

$$\frac{A}{20} = \log\left(\frac{U_2}{U_1}\right) \Rightarrow 10^{\frac{A}{20}} = U_2/U_1$$

$$U_2 = U_1 \cdot 10^{\frac{A}{20}} = 0,775 \text{ V} \cdot 10^{-3/20} = \underline{\underline{0,55 \text{ V}}}$$

alternativ $-3 \text{ dB} \rightarrow \text{Feldgröße} \cdot \frac{1}{\sqrt{2}}$

$$U_2 = U_1 \cdot \frac{1}{\sqrt{2}} = 0,775 \cdot \frac{1}{\sqrt{2}} = \underline{\underline{0,55 \text{ V}}}$$

Bsp 16 P bei 64dBc

$$P_1 = 1 \text{ mW}$$

$$A = 10 \log \left(\frac{P_2}{P_1} \right)$$

$$\frac{A}{10} = \log \left(\frac{P_2}{P_1} \right)$$

$$P_2 = P_1 \cdot 10^{\frac{A}{10}} = 1 \text{ mW} \cdot 10^{\frac{64}{10}} = \underline{\underline{2,51 \text{ W}}}$$

~~Bsp 17) a) $\frac{U_2}{U_1} = 0,001$~~

~~$$A = 20 \log \left(\frac{U_2}{U_1} \right) = 20 \log (0,001) =$$~~

~~$$\frac{A}{20} = \log \frac{U_2}{U_1}$$~~

~~$$U_2 = U_1 \cdot 10^{\frac{A}{20}} =$$~~

Bsp 17 a) $\frac{U_2}{U_1} = 0,001$

$$A_{21} = 20 \log \left(\frac{U_2}{U_1} \right) = 20 \log (0,001) = \underline{\underline{-60 \text{ dB}}}$$

$$A_{12} = 20 \log \left(\frac{U_1}{U_2} \right) = 20 \log \left(\frac{1}{0,001} \right) = \underline{\underline{60 \text{ dB}}}$$

b) $\frac{I_2}{U_1} = 40 \text{ S} \quad (R = 600 \Omega)$

$$\frac{I_2 \cdot R}{U_1} = \frac{U_2}{U_1} = 40 \cdot 600 = 24000$$

$$A_{21} = 20 \log \frac{U_2}{U_1} = 20 \log (24000) = \underline{\underline{87,6 \text{ dB}}}$$

$$A_{12} = 20 \log \frac{U_1}{U_2} = 20 \log \left(\frac{1}{24000} \right) = \underline{\underline{-87,6 \text{ dB}}}$$

c) $\frac{P_2}{P_1} = 42000 \quad (\text{Energieröße !!})$

$$A_{21} = 10 \cdot \log \left(\frac{P_2}{P_1} \right) = \underline{\underline{46,23 \text{ dB}}}$$

$$A_{12} = 10 \cdot \log \left(\frac{P_1}{P_2} \right) = 10 \log \left(\frac{1}{42000} \right) = \underline{\underline{-46,23 \text{ dB}}}$$

$$18) \quad 20 \text{ dB} \rightarrow \cdot 10$$

$$U_2 = U_1 \cdot 10 = 10 \text{ V} \cdot 10 = \underline{\underline{100 \text{ V}}}$$

$$19) \quad 17 \text{ dB} \quad U_1 = 10 \text{ V}$$

$$A = 20 \log\left(\frac{U_2}{U_1}\right)$$

~~$$U_2 = U_1 \cdot 10^{\frac{A}{20}}$$~~

$$U_2 = U_1 \cdot 10^{\frac{A}{20}} = 10 \text{ V} \cdot 10^{\frac{17}{20}} = \underline{\underline{70,8 \text{ V}}}$$

alternativ $17 \text{ dB} = 20 \text{ dB} - 3 \text{ dB}$

$$10 \text{ V} \cdot 10 \cdot \frac{1}{\sqrt{2}} = \underline{\underline{70,7 \text{ V}}}$$

$$\hookrightarrow -3 \text{ dB ist nur ca. } \frac{1}{\sqrt{2}}$$

$$20) \quad 0 \text{ dB} \Rightarrow \cdot 1$$

$$U_2 = U_1 = \underline{\underline{10 \text{ V}}}$$

$$21) \quad -3 \text{ dB} \approx \frac{1}{\sqrt{2}}$$

$$U_2 = U_1 \cdot \frac{1}{\sqrt{2}} = \underline{\underline{7,07 \text{ V}}}$$

~~22)~~

$$22) \quad U_1 = 10V \quad A = -12dB$$

$$A = 20 \log\left(\frac{U_2}{U_1}\right)$$

$$U_2 = U_1 \cdot 10^{\frac{A}{20}} = 10V \cdot 10^{-\frac{12}{20}} = \underline{\underline{2,5V}}$$

$$23) \quad U_1 = 10V \quad A = -20dB \rightarrow \cdot 0,1 \rightarrow U_2 = \underline{\underline{1V}}$$

$$24) \quad P_1 = 10W \quad A = 20dB \rightarrow \cdot 100 \rightarrow P_2 = \underline{\underline{1000W}}$$

↳ wegen P

~~$$25) \quad P_1 = 10W \quad A = 14dB$$

$$P_2 = P_1 \cdot 10^{\frac{14}{10} \leftarrow A}$$~~

$$25) \quad P_1 = 10W \quad A = 14dB$$

$$A = 10 \cdot \log\left(\frac{P_2}{P_1}\right)$$

$$P_2 = P_1 \cdot 10^{\frac{A}{10}} = 10W \cdot 10^{\frac{14}{10}} = \underline{\underline{251,2W}}$$

$$26) \quad A_{Tx} = -20 \text{ dBm} \quad P_{ref} = 1 \text{ mW} \quad (\text{da dBm})$$

$$A_{Rx} = -70 \text{ dBm}$$

$$a) \quad A_{Tx} = 10 \cdot \log \left(\frac{P_{Tx}}{P_{ref}} \right)$$

$$\bullet \quad P_{Tx} = P_{ref} \cdot 10^{\frac{A_{Tx}}{10}} = 1 \text{ mW} \cdot 10^{\frac{-20}{10}} = \underline{\underline{10 \mu\text{W}}} = 10^{-5} \text{ W}$$

$$b) \quad A_{Rx} = 10 \log \left(\frac{P_{Rx}}{P_{ref}} \right)$$

$$P_{Rx} = P_{ref} \cdot 10^{\frac{A_{Rx}}{10}} = 1 \text{ mW} \cdot 10^{\frac{-70}{10}} = \underline{\underline{100 \text{ pW}}} = 10^{-10} \text{ W}$$

$$c) \quad A = 10 \log \left(\frac{P_{Rx}}{P_{Tx}} \right) = 10 \log \left(\frac{10^{-10}}{10^{-5}} \right) = \underline{\underline{-50 \text{ dB}}}$$

alternativ:

$$a) \quad A_{Tx} = -20 \text{ dB} = 2 \cdot -10 \text{ dB} \rightarrow \cdot 0,01$$

$$P_2 = 0,01 \cdot P_1 = 10^{-2} \cdot 10^{-3} \text{ W} = \underline{\underline{10^{-5} \text{ W}}}$$

$\uparrow P_{Tx} \qquad \qquad \uparrow P_{ref}$

$$b) \quad A_{Rx} = -70 \text{ dB} = -7 \cdot 10 \text{ dB} \Rightarrow \cdot 10^{-7}$$

$$P_2 = 10^{-7} \cdot P_1 = 10^{-7} \cdot 10^{-3} = \underline{\underline{10^{-10} \text{ W}}}$$

$\uparrow P_{Rx} \qquad \qquad \uparrow P_{ref}$

1) $-20 \text{ dBm} \Rightarrow -70 \text{ dBm}$

$\hookrightarrow A = -70 - (-20) = \underline{\underline{-50 \text{ dBm}}}$

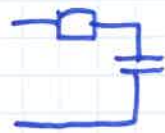
27)



$$f_g = \frac{1}{2\pi} \cdot \frac{1}{RC} = \frac{1}{2\pi} \cdot \frac{1}{100\Omega \cdot 10 \cdot 10^{-9} \text{ F}} =$$

$$= \underline{\underline{159,2 \text{ kHz}}}$$

~~28)~~ 29)



$f_g = 100 \text{ Hz}$ $R = 100 \Omega$
 $C = ?$

$$f_g = \frac{1}{2\pi} \cdot \frac{1}{RC}$$

$\Downarrow$

$$C = \frac{1}{2\pi} \cdot \frac{1}{R} \cdot \frac{1}{f_g} = \frac{1}{2\pi} \cdot \frac{1}{100\Omega} \cdot \frac{1}{100 \text{ Hz}} =$$

$$= \underline{\underline{16 \mu\text{F}}}$$



28)



$$f_g = 10 \text{ kHz} \quad R = 100 \Omega$$

$$f_g = \frac{1}{2\pi} \cdot \frac{1}{RC}$$

$$C = \frac{1}{2\pi} \cdot \frac{1}{R} \cdot \frac{1}{f_g} = \frac{1}{2\pi} \cdot \frac{1}{100 \Omega} \cdot \frac{1}{10000 \text{ Hz}} = \underline{\underline{159 \text{ nF}}}$$

30)



$$f_g = 10 \text{ kHz} \quad R = 100 \Omega$$

$$f_g = \frac{1}{2\pi} \cdot \frac{1}{RC} \Rightarrow C = \frac{1}{2\pi} \cdot \frac{1}{R} \cdot \frac{1}{f_g}$$

$$= \frac{1}{2\pi} \cdot \frac{1}{100 \Omega} \cdot \frac{1}{10000 \text{ Hz}} = \underline{\underline{159 \text{ nF}}}$$

31)



$$f_g = 100 \text{ Hz} \quad R = 100 \Omega$$

$$f_g = \frac{1}{2\pi} \cdot \frac{1}{RC} \Rightarrow C = \frac{1}{2\pi} \cdot \frac{1}{R} \cdot \frac{1}{f_g} = \frac{1}{2\pi} \cdot \frac{1}{100 \Omega} \cdot \frac{1}{100 \text{ Hz}} = \underline{\underline{16 \mu\text{F}}}$$

32)

~~29)~~

$$f_g = 100 \text{ Hz}$$

 $\downarrow -3 \text{ dB}$

$$d) U_1 = 10 \text{ V} \quad f = 100 \text{ Hz} = f_g \rightarrow \cdot \frac{1}{\sqrt{2}} \rightarrow U_2 = 10 \text{ V} \cdot \frac{1}{\sqrt{2}} = \underline{\underline{7,07 \text{ V}}}$$

$$b) U_1 = 10 \text{ V} \quad f = 10 \text{ Hz} \ll f_g \rightarrow \cdot 1 \rightarrow U_2 = U_1 = \underline{\underline{10 \text{ V}}}$$

$$c) U_1 = 10 \text{ V} \quad f = 1 \text{ kHz} = 10 \cdot f_g \rightarrow -20 \text{ dB} \rightarrow \cdot 0,01$$

$$i) U_1 = 10 \text{ V} \quad f = 1 \text{ kHz} = 10 \cdot f_g \rightarrow -20 \text{ dB} \rightarrow \cdot 0,1$$

$$U_2 = 0,1 \cdot U_1 = 0,1 \cdot 10 \text{ V} = \underline{\underline{1 \text{ V}}}$$

33)

~~30)~~

$$f_g = 100 \text{ Hz}$$

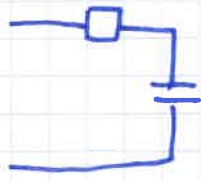
$$d) U_1 = 10 \text{ V} \quad f_1 = 100 \text{ Hz} = f_g \rightarrow \cdot \frac{1}{\sqrt{2}} \rightarrow U_2 = U_1 \cdot \frac{1}{\sqrt{2}} = \underline{\underline{7,07 \text{ V}}}$$

$$b) U_1 = 10 \text{ V} \quad f_2 = 10 \text{ Hz} = 0,1 \cdot f_g \rightarrow -20 \text{ dB} \rightarrow \cdot 0,1$$

$$U_2 = U_1 \cdot 0,1 = \underline{\underline{1 \text{ V}}}$$

$$c) U_1 = 10 \text{ V} \quad f_3 = 1 \text{ kHz} \gg f_g \rightarrow \cdot 1 \rightarrow U_2 = U_1 = \underline{\underline{10 \text{ V}}}$$

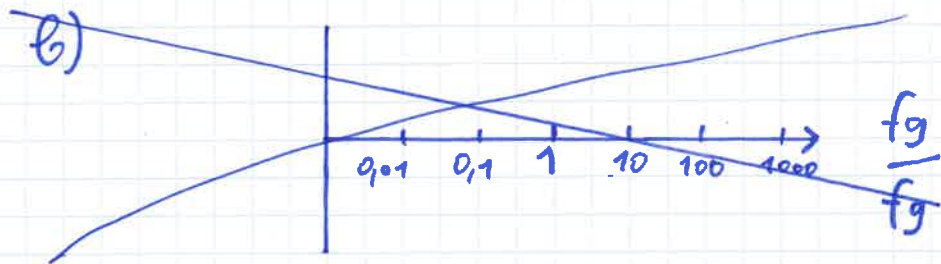
34 a)



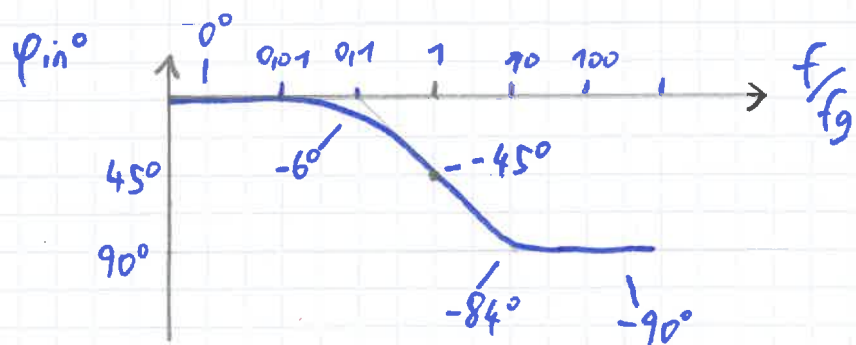
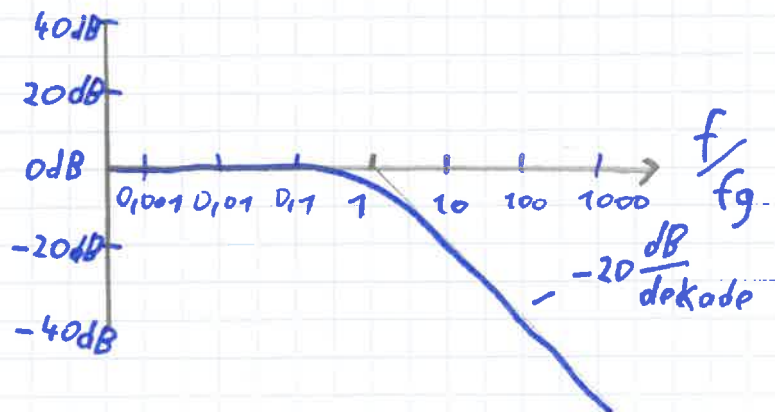
$$R_1 = 100 \text{ k}\Omega$$

$$C_1 = 15,9 \text{ nF}$$

$$f_g = \frac{1}{2\pi} \cdot \frac{1}{RC} = \frac{1}{2\pi} \cdot \frac{1}{100000 \Omega \cdot 15,9 \cdot 10^{-9} \text{ F}} = \underline{\underline{100 \text{ Hz}}}$$



$A_{in} \text{ dB}$



→ Durchlassbereich

$$i) \quad U_{1a} = 1V \quad f_{1a} = 0,1 \cdot f_g \rightarrow U_c = 1V$$

$$U_{1b} = 1V \quad f_{1b} = f_g \rightarrow U_c = \frac{1}{\sqrt{2}} \cdot 1V = \underline{\underline{0,7V}}$$

↳ bei f_g ! (-3dB)

$$U_{1c} = 1V \quad f_{1c} = 100 f_g \quad \text{im Sperrbereich}$$

$$U_c \approx 0V$$

(genau 100 $\rightarrow$ 20dB, also $\cdot 0,01 \rightarrow 0,01V$)

$$d) \quad U_{1c} = 1V \quad f_{1c} = 100 f_g \rightarrow \varphi = \underline{\underline{-90^\circ}}$$

$$e) \quad A = 20 \cdot \log\left(\frac{U_{c1}}{U_{1d}}\right)$$

$$U_{c1} = U_{1d} \cdot 10^{\frac{A}{20}} = 10V \cdot 10^{\frac{-15}{20}} = \underline{\underline{1,78V}}$$

35)

geg. $U_{\text{MESS}} \Rightarrow f_m = 100 \text{ kHz}$ $U_{\text{STÖR}} \Rightarrow f_s = 100 \text{ Hz}$ ges f_g d) U_{MESS} ungedämpft $f_g \leq 10 \text{ kHz}$ (bei $f_g \rightarrow \frac{1}{\sqrt{2}} \rightarrow 1 \text{ Dekade}$) $U_{\text{STÖR}}$ -40 dB gedämpft $\rightarrow -20 \frac{\text{dB}}{\text{Dekade}} \rightarrow 2 \text{ Dekaden}$

$$\rightarrow f_g \geq 100 \text{ Hz} \cdot 100 = 10 \text{ kHz}$$

$$\rightarrow 10 \text{ kHz} \leq f_g \leq 10 \text{ kHz}$$

$$\hookrightarrow f_g = \underline{\underline{10 \text{ kHz}}}$$

$$b) f_g = \frac{1}{2\pi} \cdot \frac{1}{RC} \rightarrow R = \frac{1}{2\pi} \cdot \frac{1}{C} \cdot \frac{1}{f_g} = \frac{1}{2\pi} \cdot \frac{1}{68 \text{ nF}} \cdot \frac{1}{10 \text{ kHz}} =$$

$$= \underline{\underline{234 \Omega}}$$

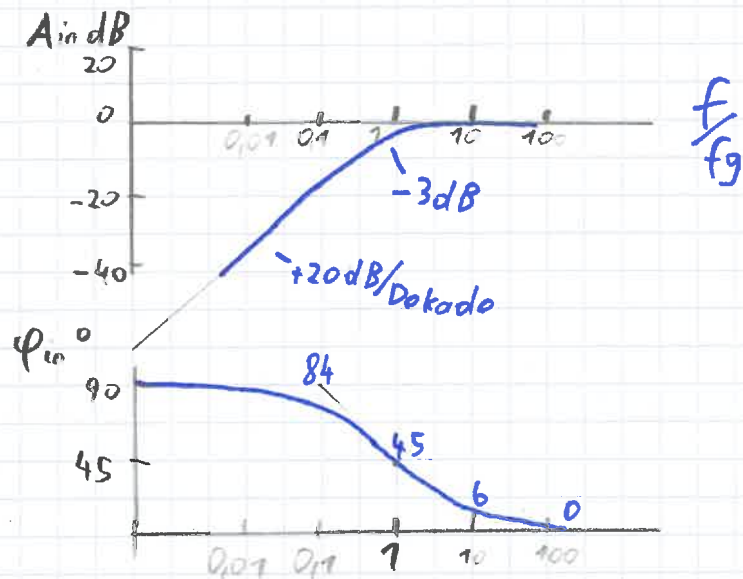
36)



→ Hochpass

$$a) \quad f_g = \frac{1}{2\pi} \cdot \frac{1}{RC} = \frac{1}{2\pi} \cdot \frac{1}{10000 \Omega \cdot 1,59 \cdot 10^{-9} \text{ F}} = \underline{\underline{10 \text{ kHz}}}$$

b)



$$i) \quad f_1 = 0,1 f_g \rightarrow -20 \text{ dB} \rightarrow \cdot 0,1 \rightarrow U_a = 0,1 \cdot U_e = \underline{\underline{0,1 \text{ V}}}$$

$$f_2 = f_g \rightarrow \frac{1}{\sqrt{2}} \rightarrow U_a = \frac{U_e}{\sqrt{2}} = \underline{\underline{0,7 \text{ V}}}$$

$$f_3 = 100 \cdot f_g \rightarrow \cdot 1 \quad (\text{Durchlassbereich}) \Rightarrow U_a = \underline{\underline{1 \text{ V}}}$$

$$d) \quad f_4 = 10 f_g \rightarrow \varphi = 6^\circ \quad (\text{ganz grob } 0^\circ)$$

$$e) \quad A = -27 \text{ dB} = 20 \cdot \log\left(\frac{U_a}{U_e}\right) \quad \text{mit } U_e = 20 \text{ V}$$

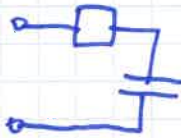
$$U_a = U_e \cdot 10^{\frac{A}{20}} = 20 \text{ V} \cdot 10^{\frac{-27}{20}} = \underline{\underline{0,9 \text{ V}}}$$

37)

$$f_{\text{MESS}} = 100 \text{ Hz}$$

$$f_{\text{STÖR}} = 100 \text{ kHz} \rightarrow \text{viel größer} \rightarrow \text{Tiefpass}$$

a)



b)

 $\rightarrow$ da ungedämpft

 $f_m:$

$$f_g \geq 10 \cdot f_m = 1 \text{ kHz}$$

 $f_s:$

$$f_g \leq 0,01 \cdot f_s = 1 \text{ kHz}$$

 $\downarrow -40 \text{ dB}$

$$1 \text{ kHz} \leq f_g \leq 1 \text{ kHz}$$

$$\rightarrow f_g = \underline{\underline{1 \text{ kHz}}}$$

c)

$$f_g = \frac{1}{2\pi} \cdot \frac{1}{R \cdot C}$$

$$R = \frac{1}{2\pi} \cdot \frac{1}{C} \cdot \frac{1}{f_g} = \frac{1}{2\pi} \cdot \frac{1}{68 \text{ nF}} \cdot \frac{1}{1000 \text{ Hz}} = \underline{\underline{2340,5 \Omega}}$$

38)

$$f_m = 100 \text{ Hz}$$

$$f_{s1} = 100 \text{ kHz}$$

$$f_{s2} = 1 \text{ MHz}$$

} viel größer $\rightarrow$ Tiefpass

a)



b)

 f_m :

$$f_g \geq 10 \cdot f_m = 1 \text{ kHz}$$

 $\rightarrow 40 \text{ dB}$ f_{s1} :

$$f_g \leq 0,01 \cdot f_{s1} = 1 \text{ kHz}$$

 f_{s2} :

$$f_g \leq 0,01 \cdot f_{s2} = 10 \text{ kHz} \rightarrow \text{irrelevant}$$

$$1 \text{ kHz} \leq f_g \leq 1 \text{ kHz} \leq 10 \text{ kHz}$$

$$\rightarrow f_g = \underline{\underline{1 \text{ kHz}}}$$

c)

$$f_g = \frac{1}{2\pi} \cdot \frac{1}{RC}$$

$$R = \frac{1}{2\pi} \cdot \frac{1}{C} \cdot \frac{1}{f_g} = \frac{1}{2\pi} \cdot \frac{1}{68 \text{ nF}} \cdot \frac{1}{1000 \text{ Hz}} = \underline{\underline{2340,5 \Omega}}$$

39



geg: $U = 24V$

$$f_r = 800 \text{ Hz}$$

$$b = 80 \text{ Hz}$$

$$I_r = 500 \text{ mA}$$

ges: R, L, C

a) $R = ?$

$$U = R \cdot I_r$$

(bei Resonanz wirkt nur R)

$$R = \frac{U}{I_r} = \frac{24V}{0,5A} = \underline{\underline{48\Omega}}$$

b) $L = ?$

$$Q = \frac{f_r}{b} = \frac{\omega_r \cdot L}{R}$$

$$L = \frac{\cancel{f_r} \cdot R}{b \cdot 2\pi \cancel{f_r}} = \frac{R}{b} \cdot \frac{1}{2\pi} = \frac{48\Omega}{80\text{Hz}} \cdot \frac{1}{2\pi} = \underline{\underline{95,5 \text{ mH}}}$$

c) $C = ?$

$$Q = \frac{f_r}{b} = \frac{1}{\omega_r \cdot C \cdot R}$$

oder $\omega_r = \frac{1}{\sqrt{LC}}$
oder $\omega_r L = \frac{1}{\omega_r C}$

$$C = \frac{1 \cdot b}{f_r^2 \cdot 2\pi \cdot R} = \underline{\underline{0,41 \mu\text{F}}} = 414 \text{ nF}$$

40)

~~39)~~

$$U = 24V$$

$$f_r = 800 \text{ Hz}$$

$$b = 80 \text{ Hz}$$

$$I_R = 0,5A$$

$$C = 414 \text{ nF}$$

$$a) \quad R = \frac{U}{I_R} = \frac{24V}{0,5A} = \underline{\underline{48 \Omega}}$$

$$b) \quad L = ? \quad Q = \frac{f_r}{b} = \frac{\omega_r \cdot L}{R} = \frac{2\pi f_r \cdot L}{R}$$

$$\frac{f_r}{b} = \frac{2\pi f_r \cdot L}{R}$$

$$L = \frac{1}{2\pi} \cdot \frac{R}{b} = \frac{1}{2\pi} \cdot \frac{48 \Omega}{80 \text{ Hz}} = \underline{\underline{95,5 \text{ mH}}}$$

$$c) \quad Q = \frac{U_L}{U} = \frac{U_C}{U}$$

↓
Resonanzüberhöhung $\approx \frac{\text{gespeicherter Energie}}{\text{Energieverlust}}$

$$Q = \frac{f_r}{b}$$

(1 Definition reicht)

$$d) \quad Q = \frac{f_r}{b} \quad Q \downarrow \Rightarrow b \uparrow \Rightarrow \text{halbieren!}$$

$$e) \quad L \quad Z_L(\omega = \infty) = j \omega L = \infty \Rightarrow U_L = 24V; U_R = 0V; U_C = 0V$$